ACADEMIC CENTRAL

PRACTICE EXAMINATION

UNIVERSITY · 3RD YEAR

AUTOMATION AND ROBOTICS

PAPER 1 · PRACTICE A · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central University · 3rd year Automation and Robotics paper 1 · practice a.

Written by Academic Central for practice, following robotics and automation modules at South African universities (e.g. University of Pretoria ERB 410 Robotics: principles, modelling and control of robotic systems), with PLC and pneumatic automation as taught in mechatronics programmes. It is not a university memorandum.

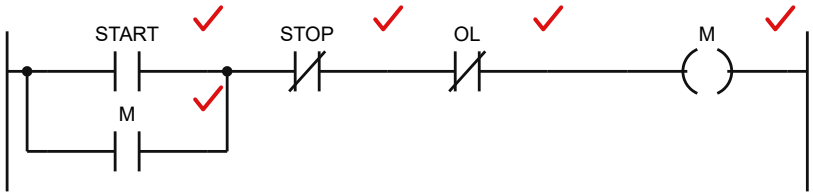
How to mark: each ✓ is one mark. Give marks for correct steps even if a later step is wrong (consistent accuracy). Accept any correct ladder program that does the job, and answers that differ only through rounding.

QUESTION 1

1.1 Read all the inputs into the input image table ✓ (3)
Run the program from top to bottom using that image ✓
Write the results to the outputs, do housekeeping and communication, then repeat ✓

1.2 Sinking input: the sensor supplies +24 V to the input and current flows into the input to 0 V (used with PNP sensors) ✓ (2)
Sourcing input: the input supplies the current, and the sensor connects it to 0 V (used with NPN sensors) ✓

1.3



START (NO) with the seal-in contact M (NO) in parallel ✓✓; STOP (NC) in series ✓; OL (NC) in series ✓; output coil M ✓ (5)

1.4 A TON timer T1 with a preset of 10 s is enabled by contact M ✓ (4)
A normally closed contact of T1 (T1.DN / T1.Q) is placed in series in the motor rung ✓, so when 10 s have passed the rung opens and M drops out ✓; the timer then resets because M is off ✓

1.5 The bottle sensor (NO) drives the count input of CTU C1 with a preset of 12 ✓ (4)
A normally closed C1 done contact in the motor rung stops the conveyor ✓
A normally open C1 done contact switches on the "box full" lamp ✓
A reset button (or the box-removed sensor) drives the counter's reset input ✓

1.6 Any TWO: function block diagram (FBD), structured text (ST), instruction list (IL), sequential function chart (SFC) ✓✓ (2)

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QUESTION 2

2.1 Single-acting: air pushes the piston one way and a spring returns it ✓ (2)
Double-acting: air pushes the piston both ways, through two ports ✓

2.2 $A = \pi \times 0.025^2 = 1.96 \times 10^{-3}\ m^2$ ✓ (3)
$F = pA = 6 \times 10^5 \times 1.96 \times 10^{-3}$ ✓ $= 1\,178.10$ N ✓

2.3 Each cylinder has limit switches (or reed sensors) at its ends, e.g. a0, a1, b0, b1 ✓ (3)
Each step is started only when the previous step's end sensor is made: A+ on start; B+ when a1 is made; A− when b1 is made; B− when a0 is made ✓
In the PLC this is written as a step sequence (SFC or set/reset bits), so steps cannot overlap ✓

2.4 5 ports and 2 positions ✓; it is moved by one solenoid and returns to its normal position by a spring when the solenoid is switched off ✓ (2)

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QUESTION 3

3.1 Degrees of freedom: the number of independent movements (joints) the robot has ✓ (6)
Workspace: all the points the end effector can reach ✓
Joints: revolute (rotating) and prismatic (sliding) ✓
SCARA: usually 4 axes, fast and stiff vertically, used for pick-and-place and assembly on a flat table ✓; articulated: 6 revolute axes, can reach any position and orientation in its workspace, used for welding, painting and handling ✓✓

3.2 $R_z(30°) = [\ \cos 30° \ -\sin 30° \ 0\ ;\ \sin 30° \ \cos 30° \ 0\ ;\ 0\ 0\ 1\ ] = [\ 0.866\ -0.5\ 0\ ;\ 0.5\ 0.866\ 0\ ;\ 0\ 0\ 1\ ]$ ✓✓ (4)
$R \times (1; 0; 0) = (0.866; 0.5; 0)$ ✓✓

3.3 $T = [\ R\ d\ ;\ 0\ 0\ 0\ 1\ ]$ with $R = R_z(90°) = [\ 0\ -1\ 0\ ;\ 1\ 0\ 0\ ;\ 0\ 0\ 1\ ]$ and $d = (2; 1; 0)$ ✓✓ (5)
$p_A = R\ p_B + d$ ✓ $= (0; 1; 0) + (2; 1; 0)$ ✓ $= (2; 2; 0)$ ✓

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QUESTION 4

4.1 $x = L1 \cos \theta1 + L2 \cos(\theta1 + \theta2)$ ✓✓ (3)
$y = L1 \sin \theta1 + L2 \sin(\theta1 + \theta2)$ ✓

4.2 $x = 0.5 \cos 30° + 0.3 \cos 90°$ ✓ $= 0.433$ m ✓ (4)
$y = 0.5 \sin 30° + 0.3 \sin 90°$ ✓ $= 0.550$ m ✓

4.3.1 $\cos \theta2 = (x^2 + y^2 - L1^2 - L2^2) / (2\ L1\ L2)$ ✓ $= (0.45 - 0.25 - 0.09) / 0.3$ ✓ $= 0.3667$ ✓ (4)
$\theta2 = 68.49°$ ✓

4.3.2 $\theta1 = \text{atan2}(y, x) - \text{atan2}(L2 \sin \theta2, L1 + L2 \cos \theta2)$ ✓ (4)
$= 26.57° - \text{atan}(0.2791 / 0.61)$ ✓ $= 26.57° - 24.59°$ ✓ $= 1.97°$ ✓

4.3.3 $\theta2 = -68.49°$ ✓ and $\theta1 = 26.57° + 24.59° = 51.16°$ ✓ (2)

4.4 A ring (annulus) centred on the base, with inner radius $L1 - L2 = 0.2$ m ✓ and outer radius $L1 + L2 = 0.8$ m ✓ (3)
(0.9; 0) is 0.9 m from the base, outside the workspace, so it cannot be reached ✓

4.5.1 $J = [\ -L1 \sin \theta1 - L2 \sin(\theta1 + \theta2) \quad -L2 \sin(\theta1 + \theta2)\ ;\ L1 \cos \theta1 + L2 \cos(\theta1 + \theta2) \quad L2 \cos(\theta1 + \theta2)\ ]$ ✓✓✓ (3)

4.5.2 $J = [\ -0.55\ -0.3\ ;\ 0.433\ 0\ ]$ ✓✓ (2)

4.5.3 $x' = -0.55 \times 0.5 - 0.3 \times 1$ ✓ $= -0.575$ m/s ✓ (3)
$y' = 0.433 \times 0.5 = 0.217$ m/s ✓

4.5.4 det J = L1 L2 sin θ2 = 0 when θ2 = 0° or 180° (arm fully stretched or folded) ✓ (2)
The arm loses a degree of freedom there: it cannot move in some directions, and the joint speeds needed become very large near it ✓

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QUESTION 5

5.1.1 θ(0) = 0 and θ′(0) = 0 give $a_0 = 0$ and $a_1 = 0$ ✓ (5)
$a_2 = 3\Delta\theta / T^2 = 3 \times 90 / 9 = 30$ ✓✓
$a_3 = -2\Delta\theta / T^3 = -180 / 27 = -6.67$ ✓
$\theta(t) = 30t^2 - 6.67t^3$ degrees ✓

5.1.2 The speed is largest halfway, at t = 1.5 s ✓: $\theta' = 60(1.5) - 20(1.5)^2 = 45$ °/s ✓ (2)

5.2 Each joint has its own controller: torque = Kp × (position error) + Kd × (speed error) ✓✓, (4)
with the joints treated as separate ✓
Gravity pulls on the links, so a PD controller alone leaves a steady position error; adding the calculated gravity torque for the present pose removes it ✓

5.3.1 v = r(ω_R + ω_L)/2 = 0.05 × 16 / 2 ✓ = 0.40 m/s ✓ (4)
ω = r(ω_R − ω_L)/L = 0.05 × 4 / 0.3 ✓ = 0.67 rad/s (turning left) ✓

5.3.2 R = v/ω = 0.4 / 0.667 ✓ = 0.6 m ✓ (2)

5.4 500 × 4 = 2 000 counts per revolution ✓✓ (3)
360° / 2 000 = 0.18° ✓

5.5 Any THREE: fences and interlocked gates; emergency-stop buttons; light curtains or (5)
safety laser scanners; safety mats; reduced speed in teach mode with an enabling switch; lockout procedures for maintenance ✓✓✓
A cobot is designed to work next to people without a fence: it limits its speed, force and power and stops when it touches something ✓✓

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TOTAL: 100

END OF MARKING GUIDELINE